Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2020-P2-Model)** **Date: 04-01-2026**

Time: 02.00Pm to 05.00Pm **GTA-10** **Max. Marks: 198**

KEY SHEET

PHYSICS

1	**0**	2	**3**	3	**6**	4	**5**	5	**4**
6	**5**	7	**AD**	8	**ABD**	9	**BD**	10	**BD**
11	**ACD**	12	**ACD**	13	**13**	14	**5.81–5.84**	15	**14**
16	**25**	17	**11**	18	**0.62**				

CHEMISTRY

19	**6**	20	**1**	21	**1**	22	**9**	23	**4**
24	**4**	25	**ABC**	26	**ABCD**	27	**ABC**	28	**BC**
29	**ABCD**	30	**AB**	31	**66**	32	**80**	33	**1791**
34	**108**	35	**4**	36	**28.32**				

MATHEMATICS

37	**2**	38	**6**	39	**5**	40	**0**	41	**2**
42	**3**	43	**ACD**	44	**AD**	45	**BCD**	46	**ABC**
47	**BCD**	48	**ABD**	49	**3**	50	**6**	51	**2**
52	**18**	53	**445**	54	**2002**				

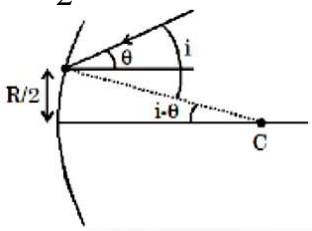
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SOLUTIONS

PHYSICS

1. The ray will be totally reflected when angle of incidence is equal to critical angle.

$$\frac{\sin i}{\sin\frac{\pi}{2}}=\frac{\mu_2}{\mu_1}\Rightarrow \sin i=\frac{\mu_2}{\mu_1}=\frac{1}{2}\Rightarrow i=30^\circ$$



From figure, $i-\theta=30^\circ\Rightarrow\theta=0$

2. $I=m\ell^2+m(2\ell)^2, \ell_C=\frac{3\ell}{2}$

$T_C=T$

$$2\pi\sqrt{\frac{I}{mg\ell_C}}=2\pi\sqrt{\frac{\ell'}{g}}\left(\frac{5m\ell^2}{mg\times\frac{3\ell}{2}}\right)=\frac{\ell'}{g}\quad \ell'=\frac{5\ell}{3}$$

$x=3$

3. Maximum possible loss of energy is when collision is perfectly inelastic is

$$(KE_{loss})_{max}=\frac{1}{2}\mu V_{rel}^2=\frac{1}{2}\times\frac{m}{2}(V-0)^2=\frac{\frac{1}{2}mV^2}{2}=\frac{25.5}{2}=12.75eV$$

This is sufficient to raise the electron upto 4th shell. While returning first shell, six photons of different wavelengths are possible.

4. For meeting, their position must be same.

$x_A=t^3+4t^2 \qquad x_B=5t^2+20t$

For meeting, $x_A=x_B\Rightarrow t^3+4t^2=5t^2+20t$

$t^3-t^2-20t=0$

$t\left[t^2-5t+4t-20\right]=0$

$t=0$, 5 and –4 s

5. 1) Calculate the net work done (W_{net})

For any reversible cycle, the net work done is the area enclosed by the cycle on the T–S diagram. The cycle $A\to B\to C\to A$ forms a right–angled triangle.

$$W_{net}=Area=\frac{1}{2}\times base\times height=\frac{1}{2}(S_0)(T_0)=\frac{1}{2}S_0T_0$$

2) Calculate the Heat Rejected $(|Q_{B-C}|)$

Heat (Q) is given by the area under the T–S curve for a given process, Q = TdS. Heat is rejected when the process involves a decrease in entropy (dS < 0).

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The magnitude of the heat rejected is: $|Q_{B-C}|$ = Area under B–C = (height)×(width) = $(2T_0)(S_0) = 2S_0T_0$.

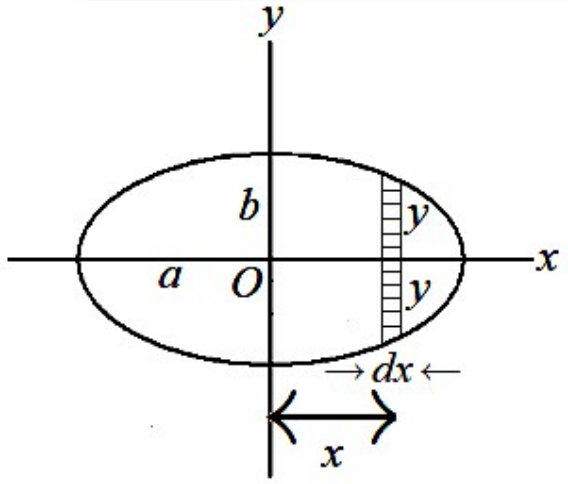
3) Find the ratio Z.

$$Z = \frac{|Q_{B\to C}|}{W_{net}} = \frac{2S_0T_0}{\frac{1}{2}S_0T_0} \Rightarrow Z = \frac{1}{1/2} = 4$$

6. From the given information, the screw gauge has negative zero error of 4 circular scale divisions. To allow 4.78 mm thick plate, the screw gauge to be unscrewed by 10 rotations.

plate cannot be inserted with 9 rotations (= 4.5 mm only). Next allowed value is 10 $\left(10\times\frac{1}{2} = 5mm\right)$

7.



$$dm = \frac{m}{\pi ab}\cdot 2y\,dx$$

$$dI_x = \frac{(dm)(2y)^2}{12} = \frac{1}{3}\cdot\frac{2mydx}{\pi ab}\cdot y^2$$

$$= \frac{2}{3}\cdot\frac{m}{\pi ab}\cdot y^3 dx \qquad \text{_____ (1)}$$

$$\frac{x^2}{a^2}+\frac{y^2}{b^2} = 1 \Rightarrow y = b\sqrt{1-\frac{x^2}{a^2}} = \frac{b}{a}\sqrt{a^2-x^2}$$

From (1), $dI_x = \frac{2m}{3\pi ab}\cdot\frac{b^3}{a^3}\left(a^2-x^2\right)^{3/2}dx$

$$I_x = 2\times\frac{2m}{3\pi ab}\cdot\frac{b^3}{a^3}\int_0^a\left(a^2-x^2\right)^{3/2}dx \qquad \text{_____ (2)}$$

Let $x = a\sin\theta \Rightarrow dx = a\cos\theta d\theta$

$x = 0 \Rightarrow \theta = 0$ and $x = a \Rightarrow \theta = \frac{\pi}{2}$

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$$\int_0^a \left(a^2-x^2\right)^{3/2} dx = \int_0^{\pi/2} a^3\cos^3\theta \cdot a\cos\theta d\theta$$

$$= a^4 \int_0^{\pi/2} \cos^4\theta d\theta = a^4 \int_0^{\pi/2} \left[\frac{1+\cos 2\theta}{2}\right]^2 d\theta$$

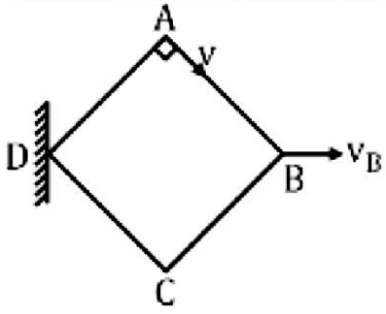
$$= \frac{a^4}{4}\left[\sin 2\theta + \frac{3}{2}\theta + \frac{1}{8}\sin 4\theta\right]_0^{\pi/2} = a^4\left(\frac{3\pi}{16}\right)$$

From (2), $I_x = \frac{4m}{3\pi ab}\cdot\frac{b^3}{a^3}\cdot a^4\cdot\frac{3\pi}{16} = \frac{mb^2}{4}$

$\|ly$ $I_y = \frac{ma^2}{4}$

8. Displacement of B, $S_B = \sqrt{2}\ell$

Velocity of B, $V_B = \sqrt{2\sqrt{2}a_B\ell} = 2^{3/4}\sqrt{a_B\ell}$

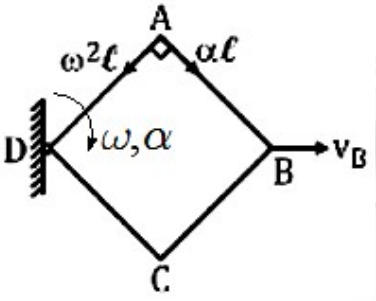


Components of velocities of A and B along AB, must be same

$\Rightarrow V_A = V_B\cos 45^\circ = 2^{3/4-1/2}\sqrt{a_B\ell} = 2^{1/4}\sqrt{a_B\ell}$

Angular velocity of DA is

$\omega = \frac{V_A}{\ell} = 2^{1/4}\sqrt{\frac{a_B}{\ell}}$



x–component of acceleration of A must be $a_{Ax} = \frac{a_B}{2}$ towards right. Acceleration of A must have two components – one along $AD\left(\omega^2\ell = \sqrt{2}a_B\right)$ and other one perpendicular to AD (say $\alpha\ell$)

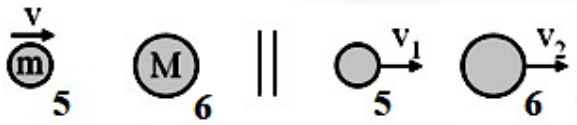
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$$\Rightarrow \frac{\alpha\ell}{\sqrt{2}} - \frac{\omega^2\ell}{\sqrt{2}} = \frac{a_B}{2}$$

$$\Rightarrow \frac{\alpha\ell}{\sqrt{2}} = \frac{3a_B}{2} \Rightarrow \alpha\ell = \frac{3}{\sqrt{2}}a_B$$

$$\therefore a_A = \sqrt{(\alpha\ell)^2 + \left(\omega^2\ell\right)^2} = \sqrt{\frac{9}{2}a_B^2 + 2a_B^2} = \left(\frac{13}{2}\right)^{1/2} a_B$$

9.



$mv = mv_1 + Mv_2$ and $v_2 - v_1 = v$

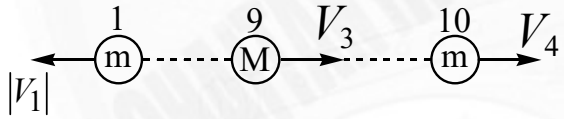
$$\therefore v_2 = \frac{2v}{\left(\frac{M}{m}+1\right)}, v_1 = \frac{m-M}{m+M}V \text{ is negative as } m < M.$$



$Mv_2 = Mv_3 + mv_4$ and $v_4 - v_3 = v_2$

$$\therefore v_4 = \frac{4Mmv}{(M+m)^2}$$

Finally:



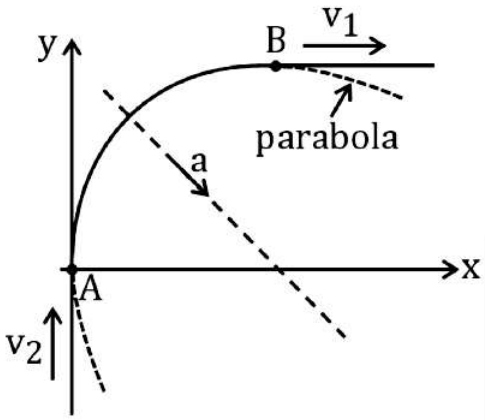
All the remaining will be at rest

10. $\Delta\vec{V} = \langle\vec{a}\rangle\Delta t$ and $\vec{a}$ is provided by f_s from ground. For a given $\Delta\vec{V}$, Δt is minimum when $a_{\max} = \mu g$ is along $\Delta\vec{V}$.

So the minimal time for his turn is $t = \frac{|\vec{v_2} - \vec{v_1}|}{\mu g} = \frac{\sqrt{2}v}{\mu g} \approx 7s$

The boy's acceleration vector is constant throughout, with magnitude μg, and directed to the southeast (it makes an angle of 45° with the x–axis), and so his trajectory during the turn (in the same way as for projectiles) is a parabola. The parabola has its axis in the northwest–southeast direction,

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Equation of path just like normal projectile is $y = v\sqrt{\frac{2\sqrt{2}}{\mu g}}\sqrt{x} - x,\ y = 5\left(2^{\frac{3}{4}}\right)\sqrt{x} - x.$

11. Current carrying charges (+ve or –ve) of the current shown experience magnetic force towards M.

12. Force on q due to dipole

$$F = \frac{1}{4\pi\varepsilon_0}\frac{2pq}{(x+y+z)^3}$$

$$(V_P + V_{ind})_0 = (V_P + V_{ind})_A$$

$$\frac{-Kp}{(x+y)^2} + \frac{Kq}{R} = \frac{-Kp}{y^2} + V_i$$

$$\therefore (V_{ind})_A = P\left(\frac{1}{y^2} - \frac{1}{(x+y)^2}\right) + \frac{q}{R}$$

13. COAM about peg

$$M\omega\ell(r-\ell) - \frac{M\ell^2\omega}{12} = 0$$

$$r - \ell = \frac{\ell}{12}$$

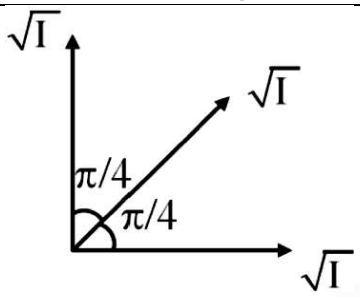
$$r = \frac{13\ell}{12}$$

14. $\Delta S_{12} = \frac{17\lambda}{4}\sin 30^\circ = \frac{17\lambda}{8}$

$$\Delta\phi_{12} = \frac{2\pi}{\lambda}\cdot\frac{17\lambda}{8} = \frac{17\pi}{4} = 4\pi + \frac{\pi}{4}$$

$$\Delta\phi_{12} = 4\pi + \frac{\pi}{4}$$

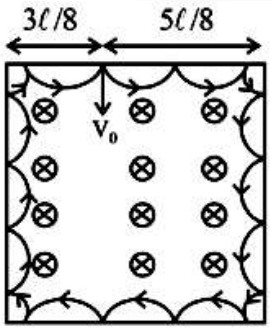
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$I_{res} = \left(\sqrt{I} + \sqrt{2I}\right)^2$

$= \left(3 + 2\sqrt{2}\right) I = 5.83I$

15.



$r = \frac{mv_0}{qB} = \frac{qB\ell}{qB8m} = \frac{\ell}{8}$

When the particle undergoes a perfectly elastic head–on collision with the walls its velocity is reversed and the centre of rotation shifts by $2r = \frac{\ell}{4}$. Furthermore, observe that near the edge of the square the particle still rotates with its centre of rotation being a vertex of the square–causing its subsequent collision with other edge of the square to also be head–on collision total number of rotation = 7.

$t = 7T = \frac{14\pi m}{qB}$

$k = 14$

16. AE – Voltage phasor across R_0

EB – Voltage phasor across L_1

AD – Voltage phasor across 'C'

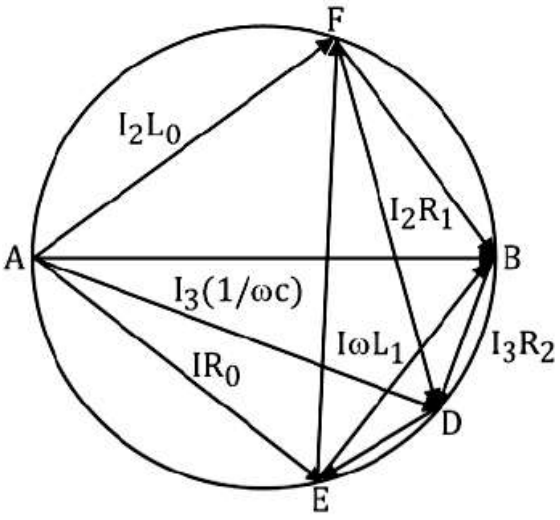
DB – Voltage phasor across R_2

AF – Voltage phasor across L_0

FB – Voltage phasor across R_1

AB – Voltage phasor across AB whose magnitude is V_0

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For ΔEDF, DE = 7V = a, DF = 15 V = b, EF = 20 V = C

By Heron's formula $S = \frac{a+b+c}{2} = 21$

Area of ΔEDF

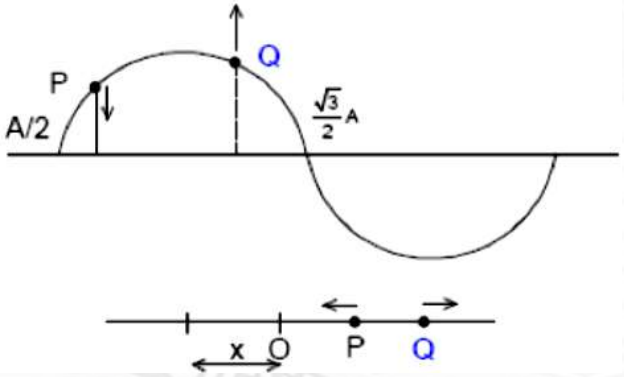
$= A = \sqrt{s(s-a)(s-b)(s-c)}$

$= \sqrt{21\times14\times6\times1} = 42$ unit

Now diameter of circle $= \frac{abc}{2A} = \frac{7\times15\times20}{2\times42} = 25$ volt

$V_A - V_B = V_0 = 25$ volt

17. In case of +ve direction proceeding of wave.



In this case PQ will be horizontal if P and Q have same position moving in opposite directions. It will be in left half at position x.

For P, $\frac{T}{12} + \frac{T}{4} + \frac{1}{\omega}\cos^{-1}\left(\frac{x}{A}\right) = t_1$

For Q, $\frac{T}{12} + \frac{T}{4} + \frac{1}{\omega}\sin^{-1}\left(\frac{x}{A}\right) = t_1$

Equating these two, we get $x = \frac{A}{\sqrt{2}}$

Hence $t_1 = \frac{11T}{24}$

In case of propagating along –ve x direction:

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For PQ to be horizontal, it will be at position in right half.

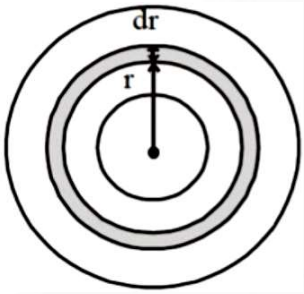
$$\frac{1}{\omega}\sin^{-1}\left(\frac{x}{A}\right)-\frac{T}{12}=\frac{1}{\omega}\cos^{-1}\left(\frac{x}{A}\right)-\frac{T}{12}$$

For P For Q

$$x=\frac{A}{\sqrt{2}} \text{ hence } t_2=\frac{T}{24}$$

$$\frac{T_1}{T_2}=11$$

18.



$$dR=\frac{\rho dr}{4\pi r^2}$$

$$R=\int dR=\frac{\rho}{4\pi}\int_{r_1}^{r_2}\frac{dr}{r^2}\Rightarrow R=\frac{\rho(r_2-r_1)}{4\pi r_1 r_2}$$

Rate of melting is max when power dissipated is sphere is max. Using maximum power transfer theorem, it happens when $R=R_{\text{int}\,ernal}$ i.e., $\frac{\rho(r_2-r_1)}{4\pi r_1 r_2}=\frac{2}{\pi}$

$$\Rightarrow \rho=\frac{8r_1r_2}{r_2-r_1}=\frac{8(200)}{10}=\frac{160}{100} \text{ (in SI)}$$

Also $\sigma=\frac{1}{\rho}=\frac{10}{16}=\frac{5}{8}$

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CHEMISTRY

19. Step 1 – NBS (2 equiv.)
Both benzylic CH_3 groups undergo radical benzylic bromination $\rightarrow CH_2Br$ (dibromide).
Both $\rightarrow CH_2Br$ convert to $-CH_2CN \rightarrow$ dinitrile.
Both nitriles hydrolyse $\rightarrow$ dicarboxylic acid (Q)
Both –COOH convert to acid chlorides.
Both acid chlorides undergo Friedel–Crafts acylation forming two new rings.
Degree of Unsaturation (DU): Two benzene rings = 8.
Two C=O groups = 2.
Two new rings = 2.
Total DU = 12 = 2x $\rightarrow$ x = 6.
Final Answer: x = 6.

20. Balanced reaction (I): $6XeF_4 + 12H_2O \rightarrow 2XeO_3 + 4Xe + 3O_2 + 24HF$
Oxidation: Oxygen in $H_2O(-2) \rightarrow O_2(0)$.
6 oxygen atoms are oxidised, each losing $2e^- \rightarrow total = 12\ mol\ e^-$.
$Per\ 1\ mol\ XeF_4 \rightarrow 12/6 = 2\ mol\ e^- lost$.
Balanced reaction (II): $2F_2 + 2NaOH \rightarrow OF_2 + 2NaF + H_2O$.
Oxygen in $OH^-(-2) \rightarrow OF_2(+2)$, losing $4e^-$ per 2 mol F_2.
Thus per 1 mol $F_2 \rightarrow 2\ mol\ e^-$ lost.
Equating electrons lost: $2 = 2y \Rightarrow y = 1$.

21. $\Delta S = n.C_{V,m}.\ln\frac{T_2}{T_1} + nR.\ln\frac{V_2}{V_1}$

$= 1\times\frac{R}{\gamma-1}\times\ln\frac{T_2}{T_1} + 1\times R\times\ln\left(\frac{T_1}{T_2}\right)^{1/n-1}$ $\left(\text{as } T.V^{n-1} = \text{Constant}\right)$

$R.\ln\frac{T_2}{T_1}\left[\frac{1}{\gamma-1} - \frac{1}{n-1}\right] = \frac{(n-\gamma)R}{(n-1)(\gamma-1)}.\ln\tau$

22. STEP1: Cyclopentadiene $+HC\ell \rightarrow 'A'$.
Cyclopentadiene undergoes electrophilic addition with $HC\ell$ to give an allylic chloride (A).
STEP2: $A + Mg \rightarrow 'B'$.
A forms a Grignard reagent: $C_5H_8 - MgC\ell$.
STEP3: $B + CO_2 \rightarrow$ acid chloride (via $SOC\ell_2$).
Grignard reagent reacts with CO_2 forming carboxylate; $SOC\ell_2$ converts it to acid chloride.
Thus B becomes cyclopentenoyl chloride.
STEP4: B + morpholine $\rightarrow 'C'$.

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Acid chloride reacts with morpholine.

STEP5: C + acetate derivative → hydrolysis → D (major).

Acyl transfer and hydrolysis form the major product D containing:

Number of C=O groups in D (x) = 2.

Number of –OH groups in D (y) = 0.

Number of chiral carbons in D (z) = 2.

Double bond equivalents (P) = 5.

(1 ring + 2 carbonyls).

Final expression: X (y + z) + P = 2(1 + 1) + 3 = 9.

23.

$2H_2$	+	O_2	→	$2H_2O$
2a		a		0
Final = 0.4a	2a – 1.6a		a – 0.8a = 0.2a	1.6a

Now, $\frac{P}{nT}=\frac{R}{V}=$ Constant $\Rightarrow \frac{P_1}{n_1T_1}=\frac{P_2}{n_2T_2}\Rightarrow \frac{4.5}{3a\times 330}=\frac{P_2}{2.2a\times 400}$ $\therefore P_2=4$ atm

24. i, ii, iii, vi

25. A: Correct — NO_2^- produces HNO_2 in acidic medium, enabling diazotisation.
B: Correct — Aromatic diazonium ions are unstable and potentially explosive in concentrated form; therefore they must be generated fresh and dilute.
C: Correct — Azo coupling proceeds by electrophilic aromatic substitution (EAS) on the activated aromatic ring.

26. All oxidises

27. 1) $BC\ell_3/\Delta$

The benzylic tertiary alcohol forms a carbocation and undergoes intramolecular Friedel–Crafts cyclisation on the arnisole ring to give the tetracyclic ether (P).

2) $P\rightarrow Q\left(1.O_3\left(1eq\right);2.Zn/AcOH\right)$

Controlled ozonolysis of the exocyclic double bond gives a dialdehyde/diketone – type compound (Q).

3) $Q\rightarrow R\left(dil.NaOH,\Delta\right)$

Intramolecular aldol condensation between the two carbonyl group produces an $\alpha,\beta-$unsaturated ketone alcohol (R), now having three new stereocentres (marked * in the official scheme).

4) $R\rightarrow S\left(H_2/Ni,2eq.\right)$

Complete catalytic hydrogenation of the C = C and C = O functions gives a saturarted tertiary benzylic alcohol (S).

5) $S\rightarrow T\left(1.TsC\ell;2.LiA\ell H_4\right)$

First, the OH group is converted into a tosylate (good leaving group); $LiA\ell H_4$ then substitutes it by hydride, giving a hydrocarbon (T) with the same carbon skeleton (no OH).

6) $T\rightarrow U\left(Se,excess,\Delta-dehydrogenation\right)$

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Selenium–mediated dehydrogenation aromatiss the ring system, giving a polycyclic aromatic ether (U).

28. B and C are undergoing condensation polymerisation.

29. All are correct

30. Michels addition and aldol.

31. Initially, $\left[Fe(CN)_6\right]^{3-}$ and $\left[Fe(CN)_6\right]^{4-}$ have equal concentrations, so

$\left[Fe(CN)_6\right]^{3-} / \left[Fe(CN)_6\right]^{4-} = 1$

By the Nernst equation: E_initial = $E^\circ = 0.059\, log\left(\left[Fe(CN)_6\right]^{3-} / \left[Fe(CN)_6\right]^{4-}\right)$

= 0.36 + 0.059 log(1)

= 0.36 V

After contamination, $\left[Fe(CN)_6\right]^{3-}$ becomes one-tenth of its initial value while $\left[Fe(CN)_6\right]^{4-}$ remains the same, so: $\left[Fe(CN)_6\right]^{3-} / \left[Fe(CN)_6\right]^{4-} = 0.1$

New cathode potential: E_new = 0.36 + 0.059 log(0.1)

$= 0.36 + 0.059 \times (-1)$

= 0.36 – 0.059

= 0.301

Decrease in cathode potential: ΔE = E_initial – E_new = 0.36 – 0.301 = 0.059 V

0.059 V = 59 mV.

32. $t = \frac{1}{K}.\ln\frac{\left[Cr^{3+}\right]_0}{\left[Cr^{3+}\right]} = \frac{1}{9\times10^{-5}s^{-1}}.\ln\frac{100}{100-80} = 1.8\times10^4$ sec = 5 hrs

33. 1. Moles of 4–methyloct–1–ene used = $\frac{2.52}{126} = 0.02\,mol$.

2. Bromination yield = 50%, so total bromides = mol.

3. Primary bromide is the major (1 in 9 : 1), so its moles = $0.01\times\frac{9}{10} = 0.009\,mol$.

4. Reaction with diethylamine followed by K_2CO_3 gives non–ionic amine product S. 5. Mass of S = 0.009 mol $\times$ molar mass = 1791 mg.

34. If reactants are taken in stoichiometric amount, then their mass ratio does not change at any stage of reaction. For 3 mole N_2, there should be 9 mole H_2.

35. i,iv,v,vii

36. 1. P = Aniline $\left(C_6H_7N\right) \rightarrow$ molecular mass = $6\times12+7\times1+14 = 93\,g/mc$.

2. Volume of P = 18.6 mL, density = 1 g/mL $\rightarrow$ mass of P = 18.6 g $\rightarrow$ moles = 18.6/93 = 0.20 mol.

3. Reaction sequence: acetanilide formation $\rightarrow$ diazotization $\rightarrow$ Sandmeyer $\left(CuBr\right)$.

4. Molar mass of Q = $6\times12+5\times1+80 = 157\,g/mol$.

5. Mass of Q = yield $\times$ moles $\times$ molar mass = $0.60\times0.20\times157 = 18.84g \approx 28.32\,g$.

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MATHEMATICS

37. $\ell_1 : \overline{r_1} = \lambda\left(\hat{i}+\hat{j}+\hat{k}\right)$ $\ell_2 : \overline{r_2} = \left(\hat{j}-\hat{k}\right)+\mu\left(\hat{i}+\hat{k}\right)$

Let the system of planes are ax + by + cz = 0 …….. (1)

It contains $\ell_1 \Rightarrow a+b+c=0$ …….. (2)

For largest possible distance between plane (1) and ℓ_2

$\Rightarrow$ The line ℓ_2 must be parallel to the plane

$\Rightarrow$ a + c = 0 …….. (3)

(2), (3) $\Rightarrow$ b = 0, a = – c

Equation of plane H_0 is x – z = 0

$$d(H_0) = \left|\frac{0+1}{\sqrt{2}}\right| = \frac{1}{\sqrt{2}}$$

38. $y = a\left(x-\frac{1}{4}\right)^2 - \frac{9}{8}$ $y = ax^2 - \frac{ax}{2} + \frac{a}{16} - \frac{9}{8}$

comparing with $y = ax^2 + bx + c$

a = a, $b = \frac{-a}{2}, c = \frac{a}{16} - \frac{9}{8}$

$$a+b+c = a+\left(\frac{-a}{2}\right)+\left(\frac{a}{16}-\frac{9}{8}\right) = \frac{9a}{16}-\frac{9}{8}$$

a + b + c is an integer $\Rightarrow$ $9a-18$ is divisible by 16

$9a = 18+16\lambda$

$\lambda = -1 \quad \Rightarrow \quad 9a = 2 \qquad a = \frac{2}{9} = \frac{p}{q}$

$\boxed{p+q=11}$

39. $\frac{\sum x_i}{40} = 30; \frac{\sum x_i^2}{40} - 30^2 = 25 \qquad \Rightarrow \sum x_i = 1200; \sum x_i^2 = 37000$

After wrong data is omitted

$\sum x_i = 1178; \sum x_i^2 = 36756;$

$\Rightarrow$ new mean $\bar{x} = \frac{1178}{38} = 31$

new variance $\sigma^2 = \frac{36756}{38} - 31^2 \qquad \Rightarrow 38\sigma^2 = 238$

40. Here none of z_i are zero and no two of z_i are equal i.

= 1, 2, 3, 4

Let f (x) = $(x-z_1)(x-z_2)(x-z_3)(x-z_4)$

$$f(x) = (x-z_2)(x-z_3)(x-z_4)+(x-z_1)(x-z_3)(x-z_4)$$
$$+(x-z_1)(x-z_2)(x-z_4))+(x-z_1)(x-z_2)(x-z_3)$$

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$f'(z_1)=(z_1-z_2)(z_1-z_3)(z_1-z_4)==\frac{4}{z_1}$

Similarly, $f'(z_2)=\frac{4}{z_2}$

$f'(z_3)=\frac{f}{z_3}, f'(z_4)=\frac{4}{z_4}$

$\Rightarrow xf'(x)=4$ has roots $z_1z_2z_3$ & z_4.

$\Rightarrow xf'(4)-4=0$ degree polynomial has leading coefficient is 4.

$\Rightarrow x'f(x)-4=4f(x)$. $\Rightarrow \frac{f'(x)}{f(x)+1}=\frac{4}{x}$

$\log(f(x)+1)=4\log x+a$, a be any constant.

$\log(f(x)+1)=\log(ax^4)$ $f(x)+1=ax^4$

$f(x)=ax^4-1$ comparing coefficient of x^4 we get a = 1.

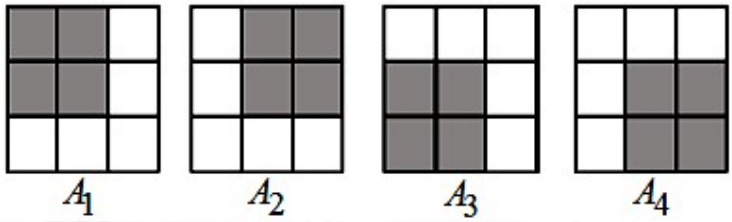
$f(x)=x^4-1$

then the roots are 1, –1, i, –i

41. We will do this by complimentary counting. Let x be the number of ways in which we have atleast one square of size, 2×2 which Red, obviously for Green also number of ways too is x.

Hence our answer = 2^9-2x.

No consider these situations.



$x=n(A_1\cup A_2\cup A_3\cup A_4)$

$=4.2^5-\{2^3+2^3+2^2+2^2+2^2+2^3+2^3\}+4.2^1-1$

= 128 – 40 + 8 – 1 = 136 – 41 = 95

Hence required answer = 512 – 190 = 322. $\Rightarrow$ $=\frac{322}{512}$

42. Since, f(x) is differentiable and hence continuous $\forall x\in R$.

$\Rightarrow f(0^+)=f(0^-)\Rightarrow P(0)=0$ and $\Rightarrow f'(0^+)=f'(0^-)\Rightarrow P'(0)=0$.

Similarly, continuity at $x=1\Rightarrow P(1)=1$ and differentiability at $x=1\Rightarrow P'(1)=0$.

Since, P(x) is a polynomial of least degree and $P'(x)$ vanishes x = 1 and x = 0.

Hence, P(x) must be cubic. $\therefore P'(x)=kx(x-1)$

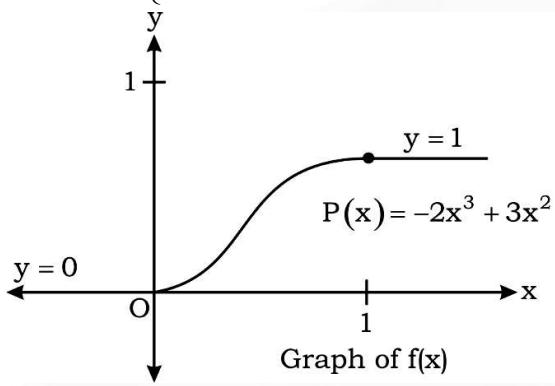
$\Rightarrow P(x)=k\left(\frac{x^3}{3}-\frac{x^2}{2}\right)+C$

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Since, $P(0)=0 \Rightarrow C=0$ and $P(1)=1 \Rightarrow k=-6$

Hence, $P(x)=-2x^3+3x^2$.

$$\therefore f(x)=\begin{cases} 0 & if \quad x\le 0 \\ -2x^3+3x^2 & if \quad 0<x<1 \\ 1 & if \quad x\ge 1 \end{cases}$$



Graph of f(x)

43. Note that $A_\alpha A_\beta = A_{\alpha+\beta}$ & $A_0 = I$

So, $A_\pi A_{\frac{3\pi}{2}} A_{\frac{5\pi}{4}} \ldots\ldots = A_S$;

Where $S=\pi+\frac{3\pi}{2}+\frac{5\pi}{2}+\ldots\ldots=\lim\limits_{n\to\infty}\pi\left(1+\frac{3}{2}+\frac{5}{4}+\ldots\ldots+\frac{2n-1}{2^n}\right)=6\pi$

So, $A_S = A_{6\pi} = I$

Further $A_{(-\alpha)} = \text{adj}(A_\alpha)$

44. $$f'(x)=\frac{(x+\cos x)(-\sin x)-(\cos x)(1-\sin x)}{(x+\cos x)^2}=\frac{-\sin x(x+\cot x)}{(x+\cos x)^2}$$

$g\left(\frac{\pi}{2}\right)=\frac{\pi}{2}$ $\qquad g(\pi)\to-\infty$

$g\left(\frac{\pi}{2}\right)g(\pi)<0$

$f'(\alpha^-)<0 f'(\alpha)>0, f'(\alpha^+)>0 \qquad \Rightarrow \alpha$ is point of minima of $f(x)$

45. Given f(f(x)) = x, put $x=0 \Rightarrow f(1)=0$

Now, $I=\int_0^1 (x-f(x))^{2026}\,dx$ …….. (i)

Put x = f(t) $\qquad \therefore dx = f'(t)dt$

So, $I=-\int_0^1 (f(t)-t)^{2026} f'(t)dt$

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$$\Rightarrow I = -\int_0^1 \left(t - f(t)\right)^{2026} f'(t)\,dt \qquad \ldots\ldots\ldots \text{(ii)}$$

(i) + (ii)

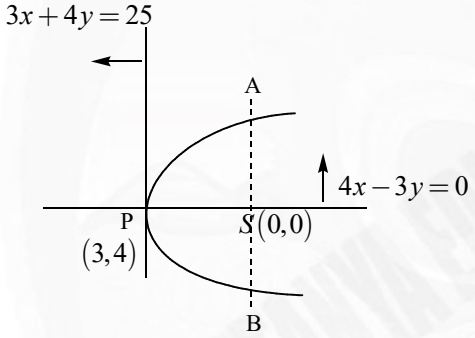
$$2I = \int_0^1 \left(t - f(t)\right)^{2026} \left(1 - f'(t)\right)dt$$

$$= \left[\frac{\left(t - f(t)\right)^{2027}}{2027}\right]_0^1 = \frac{2}{2027} \qquad \text{So, } I = \frac{1}{2027}$$

46. Slope of $SP = \frac{4}{3}$

Slope of tangent at P is $\frac{-3}{4}$ $\Rightarrow P$ is vertex of parabola

Axis of parabola is $4x - 3y = 0$

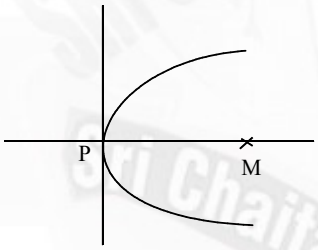


AB = length of latus reaction = 4a

a = 5

AB = 20 units

LLR = 4a = 20 units



PM = 2a, then through M only one real normal can be drawn i,o., along axis of parabola.

47. Consider $C = {}^{50}C_2 - {}^{50}C_5 + {}^{50}C_8 \ldots\ldots\ldots$

A + B + C = 0 $\qquad A + B\omega + C\omega^2 = (1-\omega)^{50}$ $\qquad A + B\omega^2 + C\omega = \left(1-\omega^2\right)^{50}$

$$\left(A + B\omega + C\omega^2\right)\left(A + B\omega^2 + C\omega\right) = (1-\omega)^{50}\left(1-\omega^2\right)^{50} = 3^{50}$$

$$\Rightarrow A^2 + B^2 + C^2 - AB - BC - CA = 3^{50}$$

$C = -A - B$ $\qquad \Rightarrow A^2 + B^2 + (A+B)^2 + (-A-B) - AB = 3^{50}$

$$\Rightarrow 3A^2 + 3B^2 + 3AB = 3^{50} \Rightarrow A^2 + B^2 + AB = 3^{49}$$ m = 3, n = 49

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48. $f(x)=xe^{-1/x}$

$f'(x)=(x)\left(e^{-\frac{1}{x}}\right)\left(\frac{1}{x^2}\right)+e^{-\frac{1}{x}}(1) \qquad =e^{\frac{-1}{x}}\left(\frac{1}{x}+1\right)$

for $x\in(0,\infty)$ $\quad f'(x)>0 \qquad \Rightarrow f(x)$ is increasing function for x > 0

$f(n)<f(n+1), n\in N \quad ne^{-1/n}<(n+1)e^{\frac{-1}{n+1}}$

$\left(\frac{n}{n+1}\right)<\frac{1}{e^{n(n+1)}}\Rightarrow\left(\frac{n}{n+1}\right)^{n(n+1)}<e$

$f(x)f\left(\frac{1}{x}\right)=\left(xe^{-1/x}\right)\left(\frac{1}{x}e^{-x}\right) \quad =e^{-\left(x+\frac{1}{x}\right)} \qquad x+\frac{1}{x}\in[2,\infty), \qquad x>0$

$\le\frac{1}{e^2}-\left(x+\frac{1}{x}\right)\in(-\infty,-2] \qquad f(x)=f'(x) \qquad \Rightarrow xe^{-1/x}=e^{-1/x}\left(\frac{1}{x}+1\right)$

$\Rightarrow x=\frac{1}{x}+1 \qquad \Rightarrow x^2=x+1 \;\; x\in(-1,0)\cup(1,2)$

49. Let $f(x)=\left|(3+\sin x)+\frac{2}{3+\sin x}+(a-3)\right|=\left|y+\frac{2}{y}+(a-3)\right|$

where y = 3 + sinx.

Let $g(y)=y+\frac{2}{y}\Rightarrow g'(y)=1-\frac{2}{y^2}>0\forall y\in[2,4]$ $\therefore$ range of $g(y)=\left[3,\frac{9}{2}\right]$

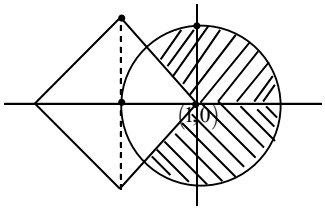
$$\therefore f(x)=\left|\underbrace{\underbrace{(3+\sin x)+\frac{2}{(3+\sin x)}}_{\le\frac{9}{2}}+(a-3)}_{\le a+\frac{3}{2}}\right| \qquad \therefore f(a)=\begin{cases}-a & \text{if } a\le-\frac{3}{4}\\ a+\frac{3}{2} & \text{if } a\ge-\frac{3}{4}\end{cases}$$

(by definition of maximum function) $\quad \therefore f(a)_{min}=\frac{3}{4}=m\Rightarrow 4m=3$

50. $(x-2)^2+(y-1)^2\le1 \qquad |x-1|+|y-1|\ge1$

Consider $x-1=x, y-1=4 \qquad (x-1)^2+y^2\le1$ and $|x|+|y|\ge1$

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Area bounded $d=\pi(1)^2-\frac{\pi(1)^2}{4}$ $\lambda=\frac{3\pi}{4}$ $4\lambda=3\pi$

51. $\sin\left(x+x^2\right)=\sin x+\sin x^2$

$2\sin\left(\frac{x+x^2}{2}\right)\cos\left(\frac{x+x^2}{2}\right)=2\sin\left(\frac{x+x^2}{2}\right)\cos\left(\frac{x-x^2}{2}\right)$

$-2\sin\left(\frac{x+x^2}{2}\right).2\sin\left(\frac{x}{2}\right)\sin\left(\frac{x^2}{2}\right)=0$ $\Rightarrow \sin\frac{x}{2}=0$ or $\sin\frac{x^2}{2}=0$ or $\sin\left(\frac{x+x^2}{2}\right)=0$

52. $x^2+y^2-8x-16y+60=0$ …….. (1)

Equation of chord of contact from (–2, 0) is $3x+4y-34=0$ …….. (2)

Solving (1) & (2) we get $5x^2-28x-12=0$ $\therefore$ Points are $\left(\frac{-2}{5},\frac{44}{5}\right)$ & (6, 4)

53. Since min(S). max(S) < 0, we must have min(S) = –a and max(S) = b for some positive integers a and b. Given a and b, there are $|S|-2=ab-2$ elements left to choose, which must come from the set $\{-a+1,-a+2.....,b-2,b-1\}$,which has size a+b–1. Therefore the number of possibilities for a given a, b are $\binom{a+b-1}{ab-2}$.

In particular, we must have $ab-2\le a+b-1\Leftrightarrow(a-1)(b-1)\le 2$. The possibilities for (a, b) to (1, n) and (n, 1) for positive integers $2\le n\le 11$(n = 1 case is impossible), and three extra possibilities: (2,2), (2,3), and 3, 2). In the first case, the number of possible sets is

$$2\left(\binom{2}{0}+\binom{3}{1}+.....+\binom{10}{8}+\binom{11}{9}\right)=2\left(\binom{2}{2}+\binom{3}{2}+.....+\binom{11}{2}\right)=2\binom{12}{3}=440$$

In the second case the number of possible sets is $\binom{3}{2}+\binom{4}{4}+\binom{4}{4}=5$.

Thus there are 445 sets in total.

54. There are fifteen 1's so the result is always a multiple of 3.
Now the number has to be a multiple of 10, so we need to insert 9 plus signs in between the fifteen 1's, so that the result is a multiple of 10.
Required number of ways = $14C_9=2002$.

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